

Step-3 is Large yet Affordable: Model-system Co-design for Cost-effective Decoding

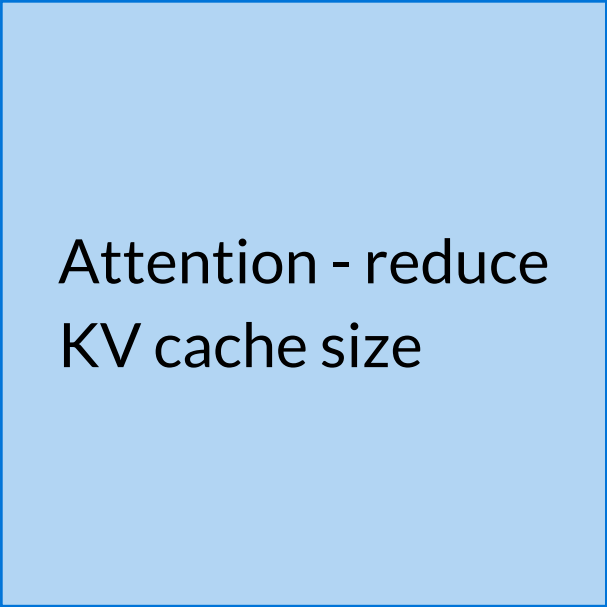
StepFun Inc.

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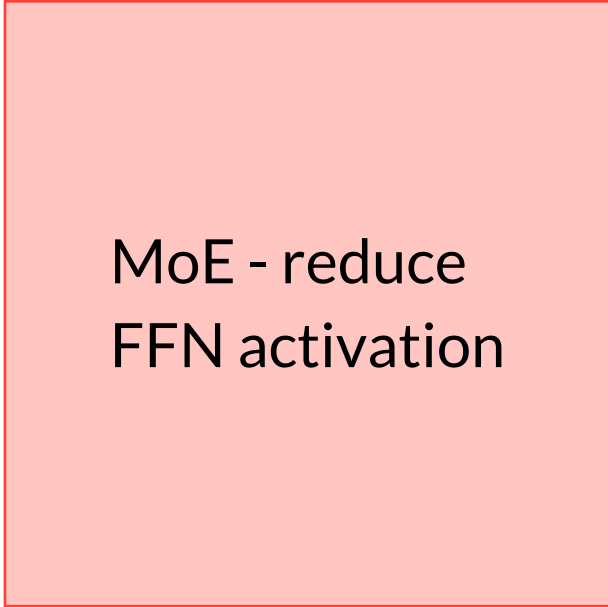
Focused Problem: Optimizing the decoding cost

Why?

- low Model FLOPs Utilization (MFU) compared with training and prefill
- reasoning model needs longer thinking
- reduce the cost for RL training
- large room for optimization



Attention - reduce
KV cache size



MoE - reduce
FFN activation

- Some work uses **excessive computational cost** to reduce the KV cache size (e.g., ??)
- Some work overly focuses on reducing the FFN activation without considering **hardware specifications**.

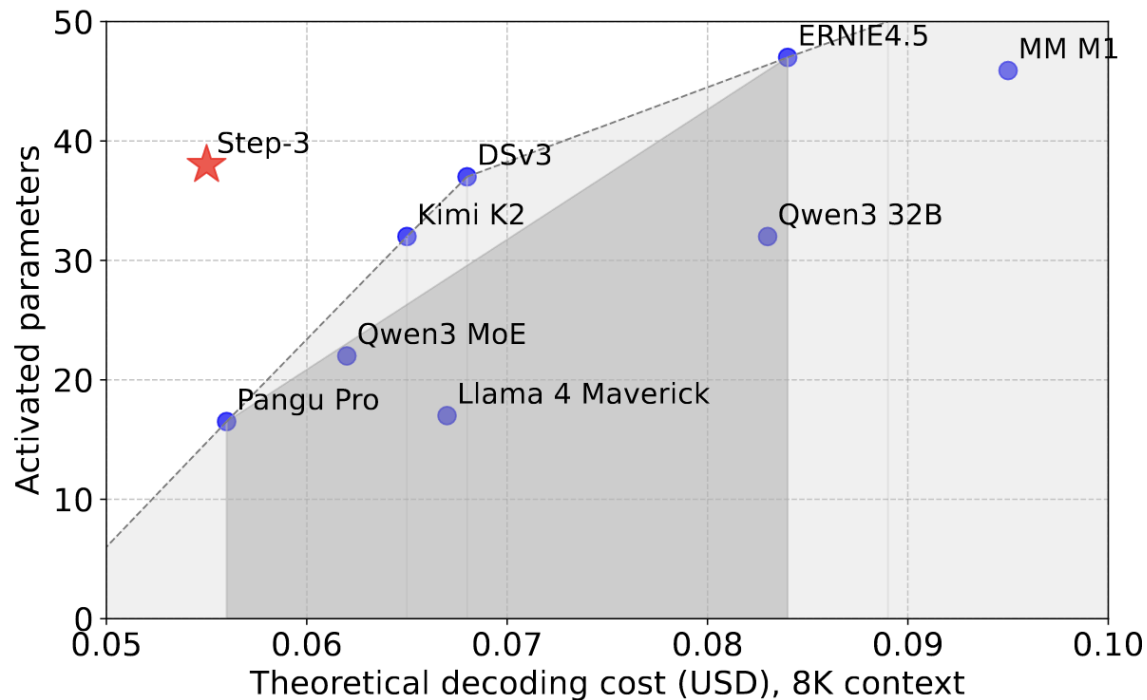


Figure 1: The Pareto frontier of recent models regarding activated parameters and decoding costs. The darker area is GQA models' Pareto frontier. Note: Step-3 also has the highest attention effective rank [7], the same as DSv3 and doubling some other models like Qwen3 MoE 235B and Kimi K2.

Helmholtz Machine

Helmholtz Machine is a hierarchical full-connected feedback NN.

RBM can be viewed as a Helmholtz machine with only **one layer of hidden units** and an **implicit posterior distribution**.

Real posterior distribution is intractable

$$p(\mathbf{h} \mid \mathbf{v}, \theta) = \frac{p(\mathbf{h}, \mathbf{v} \mid \theta)}{p(\mathbf{v} \mid \theta)}$$

Approximation with Q:

$$Q(\mathbf{h} \mid \mathbf{v}) = \prod_{i=1}^Q (h_i \mid \mathbf{v})$$

ELBO (Evidence Lower Bound) from variational inference:

$$\log p(\mathbf{v} \mid \theta) = E_Q[\log p(\mathbf{v}, \mathbf{h} \mid \theta) - \log Q(\mathbf{h} \mid \mathbf{v})] + D_{KL}(Q(\mathbf{h} \mid \mathbf{v}) \parallel p(\mathbf{h} \mid \mathbf{v}, \theta))$$

Real posterior distribution is intractable

$$D_{KL}(Q(\boldsymbol{h} \mid \boldsymbol{v}) \parallel p(\boldsymbol{h} \mid \boldsymbol{v}, \theta)) > 0$$

Therefore, the ELBO is a lower bound of the log-likelihood.

$$L(\theta, Q) = E_Q[\log p(\boldsymbol{v}, \boldsymbol{h} \mid \theta) - \log Q(\boldsymbol{h} \mid \boldsymbol{v})]$$

Modern development

Variational Autoencoders (VAEs)

Generative Adversarial Networks (GANs)

Flow-based models

Diffusion models

Works on LLM

KV-cache compression via importance evaluation

Block Attention